QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION MONTH 2025
FIFTH SEMESTER
DSE-MG5DSEBIF302-Research Methods in Biological Sciences
(2024 ADMISSION ONWARDS)

Duration: 2hrs **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Define research. [K] / [CO1]

2. What is a research problem? [K] / [CO2]

3. What is primary data? [K] / [CO1]

4.	Define frequency distribution.	[K] / [CO1]
5.	What is peer review?	[K] / [CO3]
6.	What is plagiarism?	[K] / [CO5]
7.	Name any one reference management tool.	[K] / [CO5

[2x5 = 10]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

8.	Distinguish between basic research and applied research.	[U] / [CO1]
9.	Explain the role of a hypothesis in scientific inquiry.	[U] / [CO2]
10.	State any two characteristics of a good researcher.	[U] / [CO1]
11.	Differentiate between quantitative and qualitative data.	[U] / [CO1]
12.	What is sampling? Mention any two sampling methods.	[K] / [CO1]
13.	Distinguish between single-blind and double-blind peer review.	[U] / [CO3]

14. Mention any two types of plagiarism. [K] / [CO5]

15. Explain the major components of an experimental protocol and the importance of controlling variables. [U] / [CO4]

[5x6 = 30]

Part C

Long Essay Questions

Answer any **3** questions

Each question carries **10** marks

16. Explain how charts, graphs, tables and figures can be used to communicate research findings effectively. [A] / [CO6]

17. Differentiate between descriptive statistics and inferential statistics, giving suitable examples. [An] / [CO3]

18. A researcher has collected data from a survey involving 200 participants. [An] / [CO3]

    a) Explain suitable methods for organizing and tabulating the data.
    b) Suggest appropriate charts or graphs for presenting the findings.

19. A researcher plans to investigate the effect of different concentrations of a biofertilizer on the growth of a crop. [E] / [CO4]
    a) Identify the independent, dependent and controlled variables.
    b) Outline an appropriate experimental protocol.

20. A research team is preparing a research project for publication and funding. Discuss how effective scientific communication can contribute to successful teamwork, proposal preparation, presentation of findings and publication. [E] / [CO6]

Also explain the ethical responsibilities of researchers in communicating scientific results.

[10x3 = 30]